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CS471 Project 4

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1 Introduction

This report contains the results of 120 Taillard_TestData Flow Shop Scheduling Problems (FSSP) run on NEH heuristic for both blocking and non-blocking flow shops. The results of the NEH heuristic are used to seed Ant Colony Optimization.

1.1 Flow Shop Scheduling Problem

The Flow Shop Scheduling Problem aims to reduce the time to complete n jobs by m machines. The order of jobs is permutable. With non-blocking FSSP for each job performed by a machine both the previous job on the same machine and the current job on the previous machine must first be complete. With blocking FSSP for each job exit a machine the next machine must be completely free.

1.2 NEH Heuristic

The NEH heuristic is a deterministic heuristic for FSSP that prioritizes ordering the jobs with the longest completion time first. NEH compares the two ordering for the two longest jobs and selects the permutation with the lower makespan, then it compares each job in order of longest to shortest with all possible positions to be inserted into the existing job order. Then inserting the current job into the position that leads to the best makespan. Insertions continue until all jobs are in the job order.

1.3 Ant Colony Optimization

Ant Colony Optimization is a probabilistic discrete optimization that simulates artificial ants the previous decision's of ants via evaporating virtual pheromones. In the case of FSSP the artificial ants are selecting the job order.

2 Experimentation

The results of NEH heuristic for both non-blocking and blocking flow shops can be seen in Table 3.1.
Ant Colony Optimization (ACO) was tested using the results of the completed NEH heuristic to influence the decisions of the virtual ants. Each decision was determined by both the pheromones and a heuristic view that is calculated based off the normalized total processing time of jobs. The Ant Colony Optimization was tested using the following values:

- Number of ants: 100
- Iterations: 100
- alpha: 1.0
- beta: 2.0
- evaporation: 0.5
- q: 1

The results of ACO can be seen in Table 3.1

3 Results

The results of NEH and ACO minimizing the makespan for Non-Blocking and Blocking FSSP can be seen below. On most tests ACO did find a reduced makespan from the deterministic output of ACO, however the computation time was much greater for ACO.
The improvements that ACO found in some cases was minimal or non-existent. This shows how well NEH performs on FSSP problems as a heuristic that can be applied before a meta-heuristic like ACO.

Table 3.1: NEH and ACO Makespans

	NEH				ACO			
	Non-Blocking		Blocking		Non-Blocking		Blocking	
Problem	Makespan	T(s)	Makespan	T(s)	Makespan	T(s)	Makespan	T(s)
1	1472	0.00	1701	0.00	1339	0.57	1531	0.58
2	1538	0.00	1765	0.00	1372	0.57	1530	0.57
3	1499	0.00	1655	0.00	1243	0.57	1461	0.57
4	1640	0.00	1784	0.00	1437	0.57	1677	0.56
5	1437	0.00	1558	0.00	1360	0.57	1542	0.56
6	1457	0.00	1685	0.00	1329	0.56	1588	0.56
7	1504	0.00	1706	0.00	1275	0.57	1527	0.57

8	1422	0.00	1664	0.00	1342	0.58	1605	0.56
9	1585	0.00	1641	0.00	1341	0.57	1576	0.57
10	1365	0.00	1607	0.00	1195	0.57	1453	0.56
11	2026	0.00	2203	0.00	1777	0.60	1922	0.60
12	2190	0.00	2267	0.00	1834	0.60	2032	0.60
13	1830	0.00	2026	0.00	1648	0.59	1901	0.60
14	1946	0.00	2075	0.00	1532	0.61	1733	0.61
15	1805	0.00	2027	0.00	1644	0.60	1918	0.60
16	1867	0.00	1994	0.00	1514	0.60	1776	0.61
17	1663	0.00	1826	0.00	1699	0.60	1921	0.60
18	1974	0.00	2189	0.00	1699	0.61	1943	0.61
19	1874	0.00	2080	0.00	1726	0.60	1969	0.60
20	1929	0.00	2102	0.00	1774	0.60	1995	0.60
21	2937	0.00	3012	0.00	2507	0.64	2598	0.64
22	2785	0.00	2852	0.00	2296	0.64	2501	0.65
23	2829	0.00	2935	0.00	2504	0.65	2632	0.65
24	2712	0.00	2873	0.00	2475	0.64	2662	0.64
25	2827	0.00	2958	0.00	2509	0.65	2642	0.64
26	2615	0.00	2716	0.00	2455	0.64	2618	0.65
27	2693	0.00	2831	0.00	2514	0.64	2623	0.65
28	2658	0.00	2837	0.00	2403	0.63	2560	0.65
29	2871	0.00	2968	0.00	2476	0.64	2643	0.65
30	2618	0.00	2772	0.00	2447	0.64	2579	0.65
31	3224	0.01	3997	0.01	2830	2.25	3745	2.25
32	3229	0.01	4055	0.01	2985	2.25	3885	2.25
33	3116	0.01	3773	0.01	2747	2.24	3671	2.26
34	3225	0.01	3892	0.01	2895	2.26	3750	2.26
35	3295	0.01	3976	0.01	2941	2.25	3833	2.25
36	3197	0.01	3807	0.01	3060	2.23	3832	2.24
37	3117	0.01	3764	0.01	2899	2.24	3622	2.25
38	3159	0.01	3859	0.01	2851	2.26	3755	2.25
39	3031	0.01	3604	0.01	2712	2.25	3522	2.26
40	3468	0.01	4009	0.01	2912	2.23	3764	2.24
41	3781	0.02	4478	0.02	3446	2.30	4348	2.23
42	3692	0.02	4406	0.02	3350	2.31	4116	2.33
43	3635	0.02	4349	0.02	3355	2.31	4281	2.32
44	3995	0.02	4584	0.02	3448	2.32	4350	2.32
45	3769	0.02	4404	0.02	3516	2.31	4337	2.33
46	3803	0.02	4460	0.02	3435	2.32	4268	2.33
47	3938	0.02	4517	0.02	3466	2.31	4370	2.32
48	3987	0.02	4341	0.02	3390	2.35	4203	2.34
49	3910	0.02	4512	0.02	3342	2.32	4266	2.33
50	3768	0.02	4482	0.02	3438	2.32	4237	2.33
51	4945	0.02	5429	0.03	4503	2.43	5182	2.46

52	4819	0.02	5174	0.03	4345	2.44	5007	2.45
53	4411	0.02	5197	0.03	4347	2.44	5018	2.43
4	4760	0.02	5251	0.03	4384	2.42	5055	2.45
55	4824	0.03	5349	0.03	4297	2.42	5020	2.44
56	4752	0.02	5304	0.03	4265	2.43	5013	2.43
57	4926	0.02	5418	0.03	4349	2.42	5020	2.43
58	4869	0.02	5374	0.03	4366	2.41	5052	2.43
59	4608	0.02	5253	0.03	4414	2.43	5029	2.43
60	4857	0.02	5400	0.03	4453	2.43	5140	2.44
54	4760	0.02	5251	0.03	4384	2.42	5055	2.45
55	4824	0.03	5349	0.03	4297	2.42	5020	2.44
56	4752	0.02	5304	0.03	4265	2.43	5013	2.43
57	4926	0.02	5418	0.03	4349	2.42	5020	2.43
58	4869	0.02	5374	0.03	4366	2.41	5052	2.43
59	4608	0.02	5253	0.03	4414	2.43	5029	2.43
60	4857	0.02	5400	0.03	4453	2.43	5140	2.44
61	6292	0.09	7751	0.09	5567	7.15	7679	7.17
62	5834	0.09	7359	0.09	5506	7.18	7487	7.20
63	6071	0.09	7451	0.09	5397	7.20	7271	7.19
64	5773	0.09	7185	0.09	5203	7.13	7157	7.23
65	5868	0.09	7458	0.09	5527	7.19	7414	7.20
66	5821	0.09	7458	0.09	5350	7.17	7343	7.22
67	6080	0.09	7606	0.09	5408	7.17	7493	7.23
68	6046	0.09	7368	0.10	5339	7.19	7394	7.19
69	6225	0.09	7740	0.09	5602	7.18	7643	7.16
54	4760	0.02	5251	0.03	4384	2.42	5055	2.45
55	4824	0.03	5349	0.03	4297	2.42	5020	2.44
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57	4926	0.02	5418	0.03	4349	2.42	5020	2.43
58	4869	0.02	5374	0.03	4366	2.41	5052	2.43
59	4608	0.02	5253	0.03	4414	2.43	5029	2.43
60	4857	0.02	5400	0.03	4453	2.43	5140	2.44
61	6292	0.09	7751	0.09	5567	7.15	7679	7.17
62	5834	0.09	7359	0.09	5506	7.18	7487	7.20
63	6071	0.09	7451	0.09	5397	7.20	7271	7.19
64	5773	0.09	7185	0.09	5203	7.13	7157	7.23
65	5868	0.09	7458	0.09	5527	7.19	7414	7.20
66	5821	0.09	7458	0.09	5350	7.17	7343	7.22
67	6080	0.09	7606	0.09	5408	7.17	7493	7.23
68	6046	0.09	7368	0.10	5339	7.19	7394	7.19
69	6225	0.09	7740	0.09	5602	7.18	7643	7.16
70	6140	0.09	7773	0.09	5559	7.15	7605	7.24
71	6823	0.13	8561	0.13	6416	7.28	8430	7.33
72	6215	0.12	8305	0.13	6065	7.29	8272	7.30

73	6901	0.13	8307	0.13	6206	7.28	8279	7.34
74	6775	0.12	8515	0.13	6563	7.26	8551	7.31
75	6778	0.13	8283	0.13	6186	7.27	8291	7.31
76	6394	0.12	8306	0.13	5891	7.28	8140	7.36
77	6651	0.13	8236	0.13	6159	7.29	8240	7.34
78	6735	0.12	8356	0.13	6277	7.35	8176	7.35
79	7049	0.13	8467	0.13	6378	7.26	8402	7.32
80	6479	0.12	8321	0.13	6377	7.27	8369	7.38
81	7700	0.19	9385	0.21	7166	7.44	9082	7.56
82	7714	0.19	9390	0.20	7335	7.45	9152	7.53
83	7706	0.20	9273	0.21	7293	7.52	9182	7.53
84	7713	0.19	9423	0.20	7214	7.52	9166	7.54
85	7989	0.19	9398	0.20	7316	7.47	9139	7.59
86	7770	0.19	9301	0.20	7466	7.45	9180	7.52
87	7814	0.19	9462	0.21	7399	7.45	9277	7.53
88	7994	0.19	9585	0.20	7454	7.53	9175	7.52
89	7780	0.19	9459	0.20	7413	7.42	9140	7.54
90	8206	0.19	9745	0.20	7423	7.45	9267	7.57
91	12012	0.94	16285	0.99	11631	24.38	16287	24.42
92	12412	0.94	16220	1.01	11628	24.37	16258	24.42
93	12367	0.94	16295	1.00	11803	24.32	16297	24.38
94	12098	0.93	16384	1.00	11632	24.31	16386	24.41
95	12582	0.93	16313	1.01	11602	24.31	16225	24.35
96	11977	0.93	16230	0.99	11409	24.33	16162	24.42
97	12736	0.92	16677	1.00	11749	24.29	16402	24.44
98	12098	0.95	16525	1.00	11797	24.31	16485	24.47
99	12120	0.94	16179	1.00	11500	24.35	16185	24.44
100	12190	0.93	16304	0.99	11513	24.38	16409	24.54
101	13729	1.48	17339	1.59	13015	24.76	17177	24.87
102	13601	1.47	17841	1.59	12971	24.97	17429	24.86
103	13764	1.46	17633	1.58	12969	24.80	17471	24.88
104	13600	1.47	17667	1.56	13152	24.78	17500	24.81
105	13630	1.47	17422	1.57	13038	24.93	17318	24.83
106	13748	1.48	17910	1.58	13078	24.74	17500	24.86
107	13680	1.46	17681	1.56	13182	24.72	17667	24.81
108	13820	1.46	17813	1.57	13098	24.81	17499	24.82
109	13740	1.47	17372	1.59	13049	24.77	17435	24.80
110	13682	1.46	17792	1.58	13101	24.80	17376	24.85
111	30694	22.00	42087	23.97	29185	140.37	42306	140.03
112	31435	21.83	42252	23.98	29708	141.08	42301	140.27
113	30786	21.85	42330	23.93	29355	140.37	42037	140.35
114	30546	21.68	42197	24.00	29608	140.48	42328	140.58
115	30570	21.69	41958	24.11	29364	139.88	42165	140.14
116	30072	21.70	42744	24.09	29756	140.33	42398	140.09

117	30640	21.66	42034	23.96	29392	140.10	42184	139.98
118	30804	21.58	42384	23.90	29775	139.74	42172	140.11
119	30239	21.58	42173	23.97	29323	139.51	42090	140.10
120	30070	21.93	42519	23.91	29563	140.21	42470	140.48

[1] Run on Intel Core i5 2.9 GHz, 16 GB RAM